

Absolutely — here's a **general AI Security FAQ document** suitable for awareness, customer education, or onboarding materials:

AI Security – Frequently Asked Questions (FAQ)

1. What is AI Security?

AI Security is the practice of protecting artificial intelligence systems—including models, data, and infrastructure—from malicious attacks, misuse, and failures that could lead to harmful outcomes.

2. Why is AI Security important?

AI systems increasingly make decisions in critical areas such as healthcare, finance, and cybersecurity. Without proper security, attackers can manipulate or steal models, compromise sensitive data, or cause harmful decisions.

3. What are common threats to AI systems?

Top vulnerabilities include:

- **Adversarial attacks** (crafting malicious inputs to fool models)
- **Data poisoning** (corrupting training data)
- **Model theft / extraction** (reverse engineering models)
- **Prompt injection attacks** (manipulating LLM behavior)
- **Data leakage** through outputs
- **Unauthorized access** or over-privileged use
- **Supply chain risks** in pre-trained models

4. Are GenAI and LLMs at higher risk?

Yes. LLM-based systems face unique challenges such as:

- Jailbreak attacks bypassing safety guardrails
- Toxic, unsafe, or confidential content generation
- Hallucination and trust issues
- Privacy exposure of training data

They require continuous monitoring and guardrail enforcement.

5. How do adversarial attacks work?

Attackers make small changes to an input (image, text, etc.) that are invisible to humans but cause the model to misbehave. For example, an autonomous car misreading a stop sign due to a sticker.

6. What is data poisoning?

It occurs when an attacker inserts corrupted samples into the training dataset. Later, this causes targeted errors or backdoors in the model's behavior.

7. How can organizations secure their AI models?

Key approaches:

- **Pre-deployment testing** for vulnerabilities
- **Input validation and anomaly detection**
- **Secure model endpoints and access control**
- **Guardrails for safe output generation**
- **Monitoring system behavior for deviations**
- **Regular security assessments** throughout the ML lifecycle

8. How does AI Security relate to compliance and regulations?

Regulatory frameworks such as:

- **EU AI Act**
- **NIST AI RMF**
- **ISO/IEC 42001**
include requirements for:
 - Risk assessment
 - Transparency
 - Monitoring and governance
 - Robustness and safety controls

AI Security helps ensure compliance.

9. Who is responsible for AI Security in an organization?

A cross-functional responsibility involving:

- **Data science teams**
 - **Cybersecurity teams**
 - **IT operations**
 - **Compliance & risk management**
- Security should be embedded across the entire AI lifecycle.

10. Is AI Security a one-time effort?

No. Models evolve over time due to:

- Drift in input data and behavior
 - Newly discovered attacks
 - Changing business and compliance requirements
- AI Security is a **continuous** process.

11. How is AI Security different from conventional cybersecurity?

Traditional security protects software and networks.

AI Security additionally protects:

- Model decisions
 - Training and inference pipelines
 - Model data and behavior
- Attackers specifically target how AI learns and reasons.

12. Where should organizations start?

Recommended first steps:

1. Identify critical AI assets.
2. Conduct a security and risk assessment.
3. Implement guardrails and monitoring.
4. Establish governance for responsible AI.

If you want, I can also provide:

- **A short webpage version**

- A **detailed whitepaper-style** version (15–20 FAQs)
- A **comparison** of AI Security vs. AI Safety
- **Infographics** for presentations

Would you like me to format this FAQ into a branded PDF or Word doc for AIShield awareness campaigns?